

Intellicon '26 Buildathon

Team Vortex

SCOUT

*Autonomous shelf-void detection &
micro-audit intelligence for modern retail.*

Gate 1 - Problem & Proof

Project Name	SCOUT
Team Name	Vortex
Track	Undergraduate
Domain	Retail & Supply Chain AI

Members

- G.Shazmina Oudeen – Leader
- Aathika Asmeer
- Nethmi Weerathunga
- Noorul Shabeeha

Table of Contents

Problem & Target User	2
Problem Statement	2
Target User Persona	2
Validation & Market Evidence (Route A).....	3
Research Methodology & Data Ethics	3
Primary Field Findings	3
Qualitative Evidence.....	3
International & Sri Lankan Macro Evidence.....	4
Market Opportunity	5
Sizing Summary	5
Total Addressable Market (TAM): \$2.24 Billion	5
Serviceable Addressable Market (SAM): \$165 Million	5
Serviceable Obtainable Market (SOM): \$144,000 ARR	5
Solution & Technical Feasibility	6
Proposed Solution.....	6
Why Does This Require AI?	6
Competitive Landscape.....	7
Tech Stack & Three-Week Implementation Plan.....	8
System Architecture & Technology Stack.....	8
Three-Week Implementation Roadmap	9
References.....	10
Appendix	11

Problem & Target User

Problem Statement

Modern supermarkets rely on digital inventory systems to manage stock and trigger automatic replenishment. However, the stock shown in the system does not always match what is physically available on the shelves.

Products can become unaccounted for through unrecorded shrinkage, damaged goods discarded without POS updates, or customers leaving items in the wrong aisles. These small discrepancies can create **phantom inventory** - where the system shows a product as available even though the shelf is empty.

Because the system believes stock remains, replenishment may not be triggered, allowing high-demand products to stay unavailable for days or weeks. Staff may also waste valuable time searching for products that are no longer physically available.

This leads to lost sales, abandoned purchases, customer frustration, and unnecessary staff effort. The core gap is that traditional systems track **what should be in stock**, rather than continuously reflecting **what is actually on the shelf**.

Target User Persona

Primary Users: Store Floor Supervisors and Category Replenishment Leads at multi-outlet supermarket chains such as Keells, Cargills Food City, Arpico Supercentre, and SPAR.

Role & Context: Responsible for maintaining shelf availability, coordinating restocking, and ensuring key products are available during peak hours across large stores with thousands of SKUs.

Key Pain Point: Limited real-time visibility into empty shelves. Staff often depend on manual floor checks, customer complaints, and periodic stock verification, making it easy for stockouts to go unnoticed.

Secondary Beneficiaries: Operations Managers and Category Directors who need store-level visibility into stock discrepancies to reduce stockouts and improve operational efficiency.

Validation & Market Evidence (Route A)

Research Methodology & Data Ethics

Primary research was conducted across major Sri Lankan modern trade outlets, including Cargills Food City, Keells, Arpico Supercentre, SPAR, and Glomark. **29 responses** were collected; after removing 2 non-consenting entries in line with IntelliCon guidelines, **27 participants (N = 27)** formed the validated dataset. The sample included **22 retail consumers**, **3 quick-commerce delivery pickers** from PickMe Market and Uber Eats, and **2 retail operations personnel**. All included participants provided informed consent.

Primary Field Findings

Phantom Inventory & Stockouts:

81.5% (22/27) encounter empty shelves for everyday essentials at least a few times a month, while **29.6%** experience them almost every visit. **44.4%** (12/27) have experienced phantom inventory, where POS systems showed positive stock but staff could not locate the product on the shelf or had to search the storeroom. Additionally, **66.7%** of delivery riders and app pickers reported cancellations or substitutions in at least 1 out of every 3-4 grocery orders due to unrecorded shelf stockouts.

Customer & Revenue Impact:

51.9% (14/27) leave an essential item unpurchased when their preferred brand is unavailable, while **55.6%** (15/27) estimate lost spending of **LKR 1,500–6,000+** per stockout incident. **88.9%** (24/27) would switch stores after 2-3 repeated stockouts or have already done so.

Root Causes & Operational Impact:

Participants identified damaged goods discarded without POS write-offs (**63.0%**), customer aisle dumping (**51.9%**), and backroom stock not reaching shelves (**44.4%**) as key causes. **70.4%** (19/27) observed staff spending **3-10 minutes** searching for missing items. **40.7%** reported full physical counts only quarterly, semi-annually, or after major discrepancies, allowing errors to remain for up to **90 days**. Importantly, **88.9%** (24/27) considered an automated morning checklist for identifying likely missing products valuable or extremely valuable.

Qualitative Evidence

“I went to Keells for biscuits at 9 AM; terminal showed 70 in stock, but the fridge was bare and staff said it was likely damaged earlier.”

- **Geeganage Dulma**, Supermarket Shopper

“The website showed it as available, but when I visited the store, the shelf was empty. A staff member checked and said the system stock count had not been updated after some items were sold.”

- **Basnayaka B.M.A.S**, Retail Consumer

A supermarket manager, **Ariff**, also identified customer aisle dumping and delayed backroom-to-shelf transfers as recurring issues, requiring **3-5 minutes of manual searching per incident**.

International & Sri Lankan Macro Evidence

International Literature & Theoretical Mechanisms

- **Widespread Inventory Inaccuracy:** Multi-store research found that **65% of SKU inventory records were inaccurate**, representing **28% of expected inventory value** [1]. In fast-moving grocery environments, inventory mismatch rates can exceed **60%**. Among these issues, **“phantom inventory”** - where the system shows stock even though the product is no longer physically available, is particularly problematic because it can prevent automatic replenishment [2, 7].
- **Global Revenue Impact:** The IHL Group estimates that inventory distortion costs the global retail sector around **\$1.73 trillion annually**, including **\$690.9 billion in losses from out-of-stock products** [4]. A further **\$145.2 billion is lost when store associates cannot locate products that the system shows as available** [4]. This highlights that inventory accuracy is not only a stock-management issue but also a direct revenue concern.
- **Customer Behaviour & Store-Level Causes:** Research covering **71,000 shoppers across 20 countries** found an average grocery stockout rate of **8.3%**, resulting in approximately **3.9% direct sales loss** [3, 8]. When a preferred product is unavailable, up to **31% of consumers switch to another store**. The research estimates that losing around **200 regular shoppers can represent approximately \$1 million in cumulative lost revenue** [3]. In addition, around **72% of stockouts originate from internal store-level issues**, such as delayed shelf replenishment, product misplacement, and unrecorded damaged goods, rather than upstream supplier shortages [3, 9]

Sri Lankan Modern Trade Context

- **Concentrated Retail Market:** Sri Lanka’s modern grocery sector is largely concentrated among three major supermarket chains: **Cargills Food City (~520 outlets)**, **Keells (~135 outlets)**, and **Arpico Supercentre (~65 outlets)**. Together, these operators account for over **80% of modern grocery retail volume** [5, 6].
- **Higher Stockout Rates:** Local stockout rates are estimated at **10%-14%**, nearly **1.5× the global 8.3% benchmark**. Factors such as manual stock handling, centralized distribution, and local supply-chain constraints can contribute to these discrepancies [3, 5, 10].
- **Impact on Retail Margins:** Supermarket chains operate on relatively tight margins of around **3.5%-5.5%**, according to Colombo Stock Exchange disclosures [6]. With abandoned grocery baskets potentially ranging from **LKR 1,500 to LKR 6,000+**, repeated stockouts and phantom inventory can translate into meaningful lost sales and additional operational costs [5, 6].

Market Opportunity

Sizing Summary

Market	Estimated Value	Scope
TAM	\$2.24 Billion	Global retail inventory optimization software
SAM	\$165 Million	South Asian modern grocery software market
SOM	\$144,000 ARR	Sri Lanka Year 1: 40 pilot chain stores

Total Addressable Market (TAM): \$2.24 Billion

Scope: The TAM represents the global enterprise software market covering retail inventory management, inventory distortion analytics, and out-of-stock optimization [4].

Basis: Global inventory distortion is estimated to result in approximately **\$1.73-\$1.77 trillion in annual lost revenue** [4]. Within this wider problem, the enterprise software market addressing inventory shrinkage and out-of-stock management is estimated at approximately **\$2.24 billion**, with an expected annual growth rate of **10.5%-11.5% CAGR** [4].

Serviceable Addressable Market (SAM): \$165 Million

Scope: The SAM focuses on organized modern supermarket chains across **India, Sri Lanka, Bangladesh, and Pakistan**, where inventory optimization, analytics, and automated replenishment solutions can be deployed [2, 5].

Across these four markets, there are approximately **55,000 modern trade supermarket outlets** [5]. Assuming an average annual budget of **\$3,000 per store** for inventory analytics and POS-integrated solutions [2]:

$$\text{SAM} = 55,000 \text{ stores} \times \$3,000/\text{store}/\text{year} = \$165 \text{ Million}$$

This represents the portion of the global market that is relevant to Scout's target retail environment.

Serviceable Obtainable Market (SOM): \$144,000 ARR

Scope: The Year 1 SOM focuses on **40 high-volume supermarket branches** within major urban clusters in Sri Lanka's Western Province [5, 6].

Market & Pricing Assumptions:

- Sri Lanka has approximately **850–900 chain supermarket outlets** nationwide [5, 6].
- The Year 1 target is **40 high-volume branches** across Keells and Cargills Food City, representing approximately **4.5% of modern trade outlets** [5, 6].
- The proposed SaaS subscription is **\$300 per store per month**, or **\$3,600 per store annually**.

$$\text{SOM} = 40 \text{ stores} \times \$3,600/\text{year} = \$144,000 \text{ ARR}$$

Retailer ROI: A typical metro outlet processes approximately **1,200–2,000 transactions per day**. If the system identifies just **3-4 chronic phantom-inventory SKUs**, preventing an estimated **10 abandoned baskets per day** at an average value of **LKR 2,500**, the store could recover

approximately **LKR 750,000 in monthly sales**. Compared with the proposed subscription cost of approximately **LKR 90,000 (\$300) per month**, this represents an estimated **8× return on investment** [6].

Solution & Technical Feasibility

Proposed Solution

Scout is a zero-hardware AI intelligence layer that detects phantom inventory and physical shelf voids using a supermarket's existing Point-of-Sale (POS) transaction data.

Instead of requiring expensive robotic cameras or IoT shelf sensors, Scout works with existing barcode checkout logs in real time. Its core engine uses an **unsupervised machine learning pipeline**, combining **Zero-Inflated Poisson regression** with **Isolation Forest anomaly detection** to learn normal purchasing patterns for each SKU.

When sales activity for an active product unexpectedly stops while the ERP system still reports positive inventory, Scout identifies the discrepancy and generates an **anomaly confidence score**. This helps distinguish a potential shelf void from normal changes in customer demand.

High-confidence anomalies are then processed by a lightweight **LLM reasoning agent (Gemini 1.5 Flash via free API)**. The agent connects the detected SKU to its store and aisle information and converts the findings into a prioritized **3-minute morning audit checklist**, delivered directly to floor supervisors' mobile devices.

Staff can quickly inspect the relevant shelf, confirm the issue, and reconcile the discrepancy with a single action, either requesting **immediate backroom replenishment** or recording an **unreported breakage/write-off**.

By combining statistical anomaly detection with AI-assisted task routing, Scout creates a practical feedback loop between **POS data, physical shelves, and inventory systems**. It improves visibility into shelf availability without requiring new hardware or disrupting existing checkout operations.

Why Does This Require AI?

Simple **IF/THEN rules**, such as *"flag if sales = 0 for 3 hours,"* are unreliable in grocery retail because demand changes significantly by hour, payday cycles, weather, and store footfall. Fixed thresholds can therefore generate excessive false alerts during naturally quiet periods, leading to staff alert fatigue.

Scout uses **machine learning and statistical anomaly detection** to learn the expected sales pattern of each SKU under changing store conditions. A **generative reasoning layer** then interprets these anomalies and converts them into practical, aisle-specific instructions for supervisors. This allows Scout to identify likely physical shelf voids while filtering out normal demand fluctuations without requiring additional physical sensors.

Competitive Landscape

To understand Scout's market position, the solution was compared with three common approaches to managing shelf availability: manual store checks, hardware-based shelf scanning, and traditional ERP/POS systems.

- **Manual Status Quo (Pen & Paper / Visual Checks):** Many stores still rely on staff walking the aisles, checking shelves, and responding to customer complaints. While this requires no technology investment, it is time-consuming and reactive. Staff may spend **3-10 minutes searching for missing products**, while less visible phantom stock issues can remain undetected.
- **Hardware-Based Robotics & Computer Vision (e.g., Simbe Tally, Bossa Nova):** These solutions use autonomous robots, cameras, LiDAR, and other sensors to monitor shelf conditions. They provide detailed physical visibility but require additional hardware, deployment, maintenance, and store-floor infrastructure, creating a higher implementation cost for retailers operating on tight margins.
- **Legacy ERP & Static POS Rules (e.g., SAP Retail, Oracle Retail):** These systems provide a strong foundation for inventory management but primarily depend on recorded stock balances and predefined replenishment rules. When events such as unrecorded breakage, shrinkage, or product misplacement occur, the system may continue to show stock as available even when the shelf is empty.
- **Scout:** Scout adds an **AI-powered intelligence layer** on top of existing POS data. It uses **Zero-Inflated Poisson modelling and Isolation Forest anomaly detection** to identify unusual sales patterns, then uses an **LLM reasoning agent** to turn high-confidence anomalies into prioritized, actionable audit tasks for store staff.

Competitive Comparison Matrix

Evaluation Dimension	Manual Checks	Robotic Scanners	ERP / POS Systems	Scout
Initial Investment	Low	High hardware investment	High enterprise setup	Zero CapEx SaaS model
Hardware Dependency	None	Cameras, LiDAR, robotics	Existing IT/POS infrastructure	No additional hardware
Detection Approach	Manual & reactive	Physical shelf scanning	Stock ledger & predefined rules	Transaction-based ML detection
Detection Cadence	Periodic / reactive	Scheduled scans	Dependent on recorded transactions	Near real-time anomaly monitoring
Staff Effort	High manual effort	Low-moderate	Low	~3-minute targeted audit
Intelligence	Human judgement	Computer vision	Rule-based inventory logic	ML + LLM reasoning

Phantom Inventory Detection	Limited	Possible through physical scans	Limited	Core capability
Deployment	Immediate	Hardware installation required	Enterprise integration	POS API integration
Sri Lankan Market Fit	High, but labour-intensive	Higher implementation cost	Existing but limited by ledger accuracy	Low-cost, scalable software approach
Proposed Cost	Labour cost	Hardware + service costs	Licensing/integration costs	~\$300/store/month

Tech Stack & Three-Week Implementation Plan

System Architecture & Technology Stack

Scout is designed as a **zero-hardware, low-cost software solution**, using open-source technologies and free-tier cloud services wherever possible.

Data Layer

- **PostgreSQL + Supabase:** Stores SKU information, store and aisle mappings, POS transaction events, and anomaly audit records using indexed database schemas.
- **Pandas & NumPy:** Processes incoming POS data and performs rolling sales-velocity calculations and aggregation.

Machine Learning & Anomaly Detection

- **statsmodels:** Implements **Zero-Inflated Poisson regression** to model expected purchase patterns for each SKU across different operating periods and store footfall conditions.
- **scikit-learn:** Uses **Isolation Forest** and rolling **Z-score analysis** to identify unusual sales patterns and calculate a shelf-void probability score.

AI Reasoning & Task Generation

- **Gemini 1.5 Flash:** Processes high-confidence anomaly alerts, links affected SKUs to their aisle or bay locations, and generates concise, prioritized morning audit instructions for store staff.

Backend & API

- **FastAPI + Python 3.11:** Provides the service layer for POS data ingestion, anomaly detection, audit management, and communication with the mobile interface.

Supervisor Interface

- **Next.js 14 + Tailwind CSS:** Provides a responsive **Progressive Web App (PWA)**, deployed through Vercel and optimized for supervisors using mobile devices on the shop floor.

Three-Week Implementation Roadmap

Week 1 - Gate 2: Prototype & Anomaly Detection

The first week focuses on validating the core intelligence layer.

- Develop a synthetic POS data generator covering **200+ FMCG SKUs**, including weekday/weekend sales patterns and simulated phantom-inventory scenarios.
- Build and calibrate the **Zero-Inflated Poisson** demand model and **Isolation Forest** anomaly detector.
- Set up the Supabase PostgreSQL database for SKUs, transactions, and anomaly records.
- **Deliverable:** A working ML pipeline that accepts POS data and produces verified shelf-void anomaly flags.

Week 2 - System Integration & Supervisor Interface

The second week connects the intelligence layer to the user-facing application.

- Develop FastAPI endpoints for POS ingestion, active anomaly retrieval, and audit reconciliation.
- Integrate the Gemini 1.5 Flash reasoning agent to group anomalies by aisle and generate prioritized morning audit cards.
- Build the Next.js mobile PWA with a **3-minute audit checklist** and one-tap actions: **Restock, Mark Broken/Damaged, or False Alarm.**
- **Deliverable:** An end-to-end staging system connecting the database, ML engine, AI agent, backend, and mobile interface.

Week 3 - Gate 3: Evaluation & Final Delivery

The final week focuses on testing, refinement, and demonstration readiness.

- Evaluate the model using **Precision, Recall, and False-Positive Rate** across quiet trading periods and peak weekend scenarios.
- Implement simulated ERP balance adjustments and backroom restocking workflows.
- Complete the end-to-end demonstration, presentation deck, video walkthrough, and Gate 3 documentation.
- **Deliverable:** A functional **zero-hardware shelf-void intelligence platform** ready for final evaluation and judge demonstration.

References

- [1] DeHoratius, N., & Raman, A. (2008). *Inventory Record Inaccuracy: An Empirical Analysis*. *Management Science*, 54(4), 627–641.
[DOI: 10.1287/mnsc.1070.0789](https://doi.org/10.1287/mnsc.1070.0789)
- [2] ECR Retail Loss Group (2026). *Predicting Inventory Record Inaccuracy: A Practical Guide to Smarter Counting and Phantom Stock Recovery*. ECR Retail Loss Community.
[ECR Retail Loss Research Paper](#)
- [3] Gruen, T. W., Corsten, D. S., & Bharadwaj, S. (2002). *Retail Out-of-Stocks: A Worldwide Examination of Extent, Causes and Consumer Responses*. Grocery Manufacturers of America (GMA) / CIES Food Business Forum.
[Worldwide OOS Study](#)
- [4] IHL Group (2023). "Inventory Distortion: The Good, the Bad and the Ugly of Retail Out-of-Stocks and Overstocks (\$1.77 Trillion Global Impact)." IHL Research Group / Retail TouchPoints.
<https://www.retailtouchpoints.com/features/ihl-study-inventory-distortion-will-cost-retailers-1-77-trillion-in-2023/135961/>
- [5] LMD Corporate Spotlight (2017). *Corporates Spotlight: Sri Lankan Modern Grocery Retail – Market Concentration & Chain Penetration*. Media Services Lanka / Fitch Ratings Lanka.
[LMD Corporate Spotlight Report](#)
- [6] Colombo Stock Exchange (CSE) (2024/2025). *Cargills (Ceylon) PLC & John Keells Holdings PLC Integrated Annual Reports: Retail Segment Disclosures*.
[Cargills Ceylon Annual Report](#)
[John Keells Holdings Annual Report](#)
- [7] Chuang, H. H. C., & Zhao, Y. (2014). *A Demand Estimation Approach for Inventory Inaccuracy and Phantom Out-of-Stock*. *Production and Operations Management*, 23(11), 1851–1867.
[DOI: 10.1111/poms.12198](https://doi.org/10.1111/poms.12198)
- [8] Corsten, D., & Gruen, T. (2003). *Desperately Seeking Shelf Availability: An Examination of the Extent, the Causes, and the Efforts to Address Retail Out-of-Stocks*. *International Journal of Retail & Distribution Management*, 31(12), 605–617.
[DOI: 10.1108/09590550310507731](https://doi.org/10.1108/09590550310507731)
- [9] Aastrup, J., & Kotzab, H. (2010). *Forty Years of Out-of-Stock Research: A Review and Research Agenda*. *International Review of Retail, Distribution and Consumer Research*, 20(2), 243–264.
https://www.researchgate.net/publication/233319868_Forty_Years_of_Out-of-Stock_Research-And_Shelves_Are_Still_Empty
- [10] Central Bank of Sri Lanka (CBSL). *Recent Economic Developments: Domestic Trade & Modern Retail Performance Publications*.
[CBSL Recent Economic Developments](#)

Appendix

Appendix A: Primary Research Evidence & Data Audit ($N = 27$)

- **Key Validation Metrics:** 81.5% shelf-void encounter rate; 44.4% positive-stock phantom voids; 55.6% basket loss of LKR 1,500–6,000+; 88.9% store-switching risk.
- **Raw Verification Data:** The complete anonymized survey responses (CSV, XLSX, PDF & Screenshots) and evidences are hosted in the folder:
<https://drive.google.com/drive/folders/1xDraglS1BdLT9jxPuIlnm1xmX-GsET2R>
- **Form Link :**
https://docs.google.com/forms/d/e/1FAIpQLSe7vBYrNhBBYDLEnW5J6YOWwRwvvBmPN TbZ_3CL32YrRQHLWg/viewform